Summary

We construct effective field theories of the quantum skyrmion Hall effect from matrix Chern-Simons theory for N electrons. We first consider a quantum Hall droplet within finiteN matrix Chern Simons theory. We then generalize the quantization procedure by replacing the Poisson bracket, a classical Lie derivative, with a quantum counterpart, the Lie derivative for a deformed fuzzy two-sphere. This yields a field theory of the Hall effect for a partially-filled fuzzy sphere, corresponding to matrix dimension N.

Key Findings

Microscopic field theories of the quantum skyrmion Hall effect. Extract key findings as a bulleted list, including quantitative results. We construct effective field theories from matrix Chern-Simons theory for N electrons, corresponding to matrix dimension N.